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1 import network
2 import time
3 import dht
4 import machine
5 import urequests
6 WIFI_SSID = "Wokwi-GUEST"
7 WIFI_PASSWORD = ""
8 THINGSPEAK_API_KEY = "OMWZUMQ7L9N3KZC8"
9 sensor = dht.DHT22(machine.Pin(15))
10 print("Connecting to WiFi...")
11 wifi = network.WLAN(network.STA_IF)
12 wifi.active(True)
13 wifi.connect(WIFI_SSID, WIFI_PASSWORD)
14 while not wifi.isconnected():
15     print(".", end="")
16     time.sleep(1)
17 print("\nWiFi Connected!")
18 print("IP Address:", wifi.ifconfig()[0])
19 while True:
20     try:
21         sensor.measure()
22         temperature = sensor.temperature()
23         humidity = sensor.humidity()
24         print("\nTemperature:", temperature, "°C")
25         print("Humidity:", humidity, "%")
26         url = (
27             "http://api.thingspeak.com/update"
28             "?api_key=" + THINGSPEAK_API_KEY +
29             "&field1=" + str(temperature) +
30             "&field2=" + str(humidity)
31         )
32         response = urequests.get(url)
33         print("ThingSpeak Response:", response.text)
34         response.close()
35     except Exception as e:
36         print("Error:", e)
37     time.sleep(20)
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